

University of New Brunswick
Department of Electrical and Computer Engineering
ECE 4323 – Modern Control Systems
Course Project Description

*Probabilistic Field Coverage & Autonomous Navigation for Flying Quadrotors with
Camera Measurement Uncertainties*

Instructor: Howard Li, PhD, PEng | Office: D61, Gillin Hall | E-mail: howard@unb.ca

1. Introduction

Students enrolled in ECE 4323 (Modern Control Systems) will complete a comprehensive course project. A final report and code repository are to be submitted at the end of the term. Students are expected to strictly follow the template, guidelines, and marking scheme provided by the course instructor.

The course project considers fundamental problems in mathematical modeling, non-linear state estimation, camera probabilistic sensor modeling, nested-loop flight control, and systems-level integration for autonomous flying platforms. Students will collaborate in teams to implement, simulate, and validate advanced guidance, navigation, and control (GNC) algorithms to navigate a quadrotor aerial vehicle over an uncertain workspace.

The course project emphasizes hands-on software development and flight control theory. Key technical skills practiced include:

- Quadrotor non-linear kinematics, Newton-Euler rigid-body dynamics, and aerodynamic thrust models.
- Nested-loop control system architecture (inner attitude/altitude loop & outer position/guidance loop).
- Camera vision sensor modeling with altitude-dependent field of view (FOV) and pixel resolution degradation.
- Stochastic representations of the environment via 2D spatial discretization (grid cell workspace).
- Non-linear probabilistic state estimation (Extended Kalman Filter / Particle Filter) for flying platform drift.
- Information-theoretic path planning (Shannon entropy reduction) for autonomous coverage under uncertainty.
- System integration, simulation, debugging, and formal technical report writing in Python/Matlab.

Learning Objectives

Control Abstractions: Do students master the core mathematical and physical abstractions that comprise modern multi-loop autonomous flight control systems?

System Integration: Have students developed practical skills to implement, tune, and integrate non-linear state estimators, vision sensor models, and dynamic controllers for real-world robotic challenges?

Engineering Practice: Have students demonstrated proficiency in algorithm debugging, statistical evaluation, team collaboration, technical report writing, and oral presentation?

2. Quadrotor System Dynamics & Control Architecture

The aerial platform utilized in this project is a four-rotor vertical takeoff and landing (VTOL) quadrotor UAV. The aircraft is actuated by four independent rotors arranged in a cross configuration. Two opposing rotors rotate clockwise (CW) while the other pair rotates counter-clockwise (CCW) to balance anti-torque moments.

2.1 Kinematics & Reference Frames

To transform vectors between the quadrotor Body Frame (x, y, z) and the North-East-Down (NED) Inertial Navigation Frame, the directional cosine transformation matrix R is defined using the Z-Y-X Euler angle rotation sequence (Yaw ψ , Pitch θ , Roll ϕ):

$$R = \begin{bmatrix} \cos(\psi)\cos(\theta) & \cos(\psi)\sin(\theta)\sin(\phi) - \cos(\phi)\sin(\psi) & \sin(\phi)\sin(\psi) + \cos(\phi)\cos(\psi)\sin(\theta) \\ \cos(\theta)\sin(\psi) & \cos(\phi)\cos(\psi) + \sin(\phi)\sin(\psi)\sin(\theta) & -\cos(\psi)\sin(\phi) + \cos(\phi)\sin(\psi)\sin(\theta) \\ -\sin(\theta) & \cos(\theta)\sin(\phi) & \cos(\theta)\cos(\phi) \end{bmatrix}$$

The transformation relating angular body rates $\omega = [p, q, r]^T$ to Euler angle rates $\dot{\eta} = [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$ is given by $\omega = R_r \dot{\eta}$. Around hover conditions, small angle approximations yield $R_r \approx I_{3 \times 3}$.

2.2 Rigid-Body Newton-Euler Dynamics

The rotational equations of motion derived in the body frame via Newton-Euler mechanics are:

Rotational Dynamics: $J\dot{\omega} + \omega \times (J\omega) + \omega \times [0, 0, J_r \Omega_r]^T = M_B$

where $J = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal inertia matrix, J_r is rotor moment of inertia, and $\Omega_r = -\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4$ is the overall residual rotor speed. Aerodynamic forces F_i and moments M_i produced by rotor i are proportional to the square of rotor speed Ω_i ($F_i = k_F \Omega_i^2$, $M_i = k_M \Omega_i^2$).

The translational equations of motion derived in the NED inertial frame via Newton's Second Law are:

Translational Dynamics: $m\ddot{r} = [0, 0, mg]^T + RF_B$

where $r = [x, y, z]^T$ represents the quadrotor's inertial position, m is mass, $g = 9.81 \text{ m/s}^2$, and $F_B = [0, 0, -U_1]^T$ is total vertical thrust in body coordinates.

2.3 State-Space Representation & System Parameters

The quadrotor state vector is defined as $X = [\phi, \dot{\phi}, \theta, \dot{\theta}, \psi, \dot{\psi}, z, \dot{z}, x, \dot{x}, y, \dot{y}]^T$. The control input vector $U = [U_1, U_2, U_3, U_4]^T$ maps to total thrust, roll moment, pitch moment, and yaw moment respectively.

Parameter Symbol	Physical Description	Numerical Value & Units
m	Total Aircraft Mass	0.65 kg

g	Gravitational Acceleration	9.81 m/s ²
l	Rotor Moment Arm Length	0.225 m
I_{xx}, I_{yy}	Roll & Pitch Moments of Inertia	0.0086 kg·m ²
I_{zz}	Yaw Moment of Inertia	0.0172 kg·m ²
J_r	Rotor Rotational Inertia	3.7404×10^{-5} kg·m ²
k_F	Thrust Coefficient	3.13×10^{-5} N/(rad/s) ²
k_M	Moment / Drag Coefficient	9.00×10^{-7} N·m/(rad/s) ²

2.4 Multi-Loop Control Architecture & Target Guidance

A nested-loop control architecture is implemented to decouple quadrotor control:

Inner Loop Controller: Stabilizes rapid vehicle attitude (ϕ, θ, ψ) and altitude z via proportional-derivative (PD) action, computing required control inputs U_1, U_2, U_3, U_4 .

Outer Loop Position Controller: Generates reference roll ϕ_d and pitch θ_d angles via position error PD control with a saturation function $\text{sat}(v) = \min(v_{\max}, \max(-v_{\max}, v))$ to bound tilt angles near hover.

3. Project Topic: Downward-Camera Coverage Under Uncertainty

In this project, students replace static/mobile sonar coverage models with downward-facing optical camera systems mounted on flying quadrotors. The primary objective is to survey a ground workspace W , reconstruct a probabilistic visual confidence map, and execute real-time path planning that mitigates navigation drift and vision measurement noise.

Step 1: Discretization of the 2D Ground Workspace

The continuous rectangular ground workspace W is discretized into N uniform grid cells $c^i = [x^i, y^i]^T$ for $i = 1, \dots, N$. The cell dimensions are chosen small enough such that optical coverage and illumination over any individual cell c^i can be treated as spatially uniform.

Step 2: Downward Camera Probabilistic Sensor Model

Unlike sonar sensors that emit lateral acoustic cones, a downward camera projects a rectangular optical Footprint (Field of View - FOV) onto the ground plane.

Altitude-Dependent FOV: The ground footprint dimensions scale linearly with quadrotor altitude z_t : $\text{Field}_x = 2z_t \tan(\text{FOV}_x/2)$ and $\text{Field}_y = 2z_t \tan(\text{FOV}_y/2)$. A cell c^i is within view if its relative position $[x_t^i, y_t^i]^T$ in the camera body frame lies inside the optical footprint boundary.

Probabilistic Visual Confidence Model: Instead of binary detection, visual detection confidence $P(c^i)$ follows a non-linear probabilistic function dependent on pixel resolution and optic angle off center. Detection confidence peaks at camera boresight ($x = 0, y = 0$) and degrades toward the FOV edges, while higher flying altitude increases footprint area but reduces ground pixel resolution:

$$P(c^i) = P_{\max} \exp \left(- \left((x_t^i)^2 + (y_t^i)^2 \right) / (2\sigma_{\text{cam}}^2) \right) \left(z_{\text{ref}} / (z_{\text{ref}} + |z_t - z_{\text{ref}}|) \right)$$

Step 3: Propagating Quadrotor Flight Pose Uncertainty to Coverage

Due to wind gusts, IMU noise, and GPS drift/absence, the quadrotor's 3D pose state $X_t = [x_t, y_t, z_t, \phi_t, \theta_t, \psi_t]^T$ is stochastically uncertain. The system maintains a filtered posterior pose belief $\text{bel}(X_t)$ using an Extended Kalman Filter (EKF) or Particle Filter (PF):

The uncertain location of cell c^i in the camera frame is a multivariate random variable C_t^i . By projecting the pose distribution through the camera projection geometry H , each cell maintains a probability distribution $p(W_t^i = w)$ over possible coverage levels $w \in [0.5, 1.0]$.

Step 4: Recursive Information Fusion & Confidence Map

As the quadrotor flies multiple passes over the workspace, vision observations taken at different times are fused recursively:

Maximum Function Fusion: Subsequent camera views of the same ground location are treated as statistically dependent. Fusing multiple observations utilizes a maximum function on random variables: $W_t^i = \max(W_{t-1}^i, W_t^i)$. This guarantees that cell confidence never drops below the most reliable historical observation.

Global Confidence Map: Combining all camera frames generates a global Confidence Map where each cell c^i contains an expected confidence value $p(T_t^i = 1) = E[W_t^i]$ indicating correct visual detection/classification.

Step 5: Information-Theoretic Path Planning (Shannon Entropy Reduction)

To proactively direct the quadrotor toward high-uncertainty regions, students will implement an online information-based path planner:

Entropy Reduction Objective: The planner computes expected Shannon entropy reduction $\Delta H(T_t^i)$ over the cell confidence variables. Trajectory waypoints or lawn-mower track spacings d are optimized in real time to maximize total information gain: $B(d) = \int \Delta H(T_t | \tau_d(s)) ds$.

Adaptive Track Re-planning: The planner automatically closes gaps caused by previous EKF state drift or turbulence by placing subsequent flight tracks closer together in unmapped or low-confidence areas.

Step 6: Probabilistic Mission Completion Criteria

Mission completion is evaluated probabilistically rather than by simple geometric percentage:

Threshold Criterion: A specified proportion P_1 of workspace cells must achieve a confidence probability above w_{thresh} with high certainty P_2 : $(1/N)\#\{i|p(W_t^i \geq w_{thresh}) \geq P_2\} > P_1$.

Average Confidence Criterion: Alternatively, the overall expected average confidence across all workspace grid cells must exceed a desired target w_{avg} : $(1/N)\sum E[W_t^i] > w_{avg}$ (e.g., $w_{avg} = 0.95$ or 95%).

4. Project Milestones & Marking Scheme

Milestone Phase	Deadline / Target	Key Deliverables & Tasks	Grade Weight
Milestone 1: Setup	Oct 7	Task 1: Discretization of workspace. Oral demo & presentation.	20%
Milestone 2: Basic Operation	Nov 18	Tasks 2, 3, & 4: Non-linear quadrotor state-space simulation setup. Downward camera model, EKF pose uncertainty propagation, recursive max-fusion, and intermediate report.	20%
Milestone 3: Final Demo	Dec 9	Tasks 5 & 6: Information path planning, mission completion evaluation, moving target interception demo, & presentation.	20%
Milestone 4: Final Report	Dec 21	Comprehensive final technical project report (in lieu of take-home exam) covering theory, software implementation, and results.	40%

5. Presentation & Report Guidelines

5.1 Oral Presentation & Live Demonstration

Teams will present their design and demonstrate simulation results in class (10-minute presentation followed by Q&A):

Presentation Content: Evaluated on clarity of key ideas, methodology rigor, originality, mathematical correctness, and supporting evidence (plots/animations).

Organization & Delivery: Professional delivery, equal team participation, slide visual design, clear vocal pace, and effective response to technical questions.

Live Demonstration: Demonstrate closed-loop quadrotor stability, EKF convergence, dynamic visual footprint coverage, and adaptive path re-planning.

5.2 Required Structure of the Final Written Report

Reports must adhere strictly to standard academic engineering formatting and include the following sections:

- **Title Page & Executive Summary:** Includes project title, student names, course code, instructor, and concise Abstract with keywords.
- **Table of Contents & Lists:** Formal Table of Contents, List of Figures, and List of Tables.
- **Section 1: Introduction & Problem Statement:** Formulate the aerial survey problem, camera sensing under uncertainty, and mission objectives.
- **Section 2: Background & Literature Survey:** Review state-of-the-art in quadrotor control, EKF state estimation, and probabilistic coverage.
- **Section 3: Methodology & Mathematical Derivations:** Detail quadrotor dynamics, nested PD controllers, camera FOV projection, EKF equations, and entropy planning.
- **Section 4: Simulation Experiments & Implementation:** Provide software architecture, Python scripts, test scenarios, quadrotor step responses, and coverage maps.
- **Section 5: Results & Discussion:** Critically analyze trajectory tracking, control saturation, estimation error bounds, and coverage convergence.
- **Section 6: Conclusion & Future Work:** Summarize key findings, controller performance, and potential extensions (e.g., multi-UAV swarms).
- **References & Appendices:** Complete IEEE-style references and annotated source code appendices.

6. Academic Integrity & Student Support Services

6.1 Academic Integrity Policy

The University of New Brunswick places a high value on academic integrity. Plagiarism, cheating, or unauthorized collaboration on project code and reports will not be tolerated. Penalties range from a mark of F (zero) in the course to suspension or expulsion from the University. Refer to UNB Undergraduate Calendar, Section B, Regulation VIII.A for full details.

6.2 Student Accessibility & Support Services

Students with documented disabilities requiring accommodations are encouraged to register with the UNBF Student Accessibility Centre (SAC). Additionally, the UNB Writing and Study Skills Centre and Learning Strategist Support are available to assist students with report preparation and learning techniques.

6.3 Equity, Diversity, and Inclusion (EDI)

UNB is committed to creating an inclusive intellectual community enriched by diversity. All students are respected regardless of background, gender, race, culture, or ability. Suggestions to improve class environment and inclusivity are always welcomed.